SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME:DATA ANALYSIS WITH SPSS/SAS/R

PRACTICAL 7

AIM: Selecting and dropping variables using select() in R. import dataset.

>COMMANDS USED

library(dplyr) # select() is part of the dplyr package

 # ---

 # 1. IMPORT DATASET

 # ---

```
# Import the CSV file
ai_jobs <-
read.csv("AI_Impact_on_Jobs_2030.csv
")

print("--- Original Dataset (First 3 rows)
---")
print(head(ai_jobs, 3))

#
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# 2. SELECTING VARIABLES (Keeping
Columns)
```

```
# ---------------------------------------------------------------------------------

# Method A: Select specific columns by name
# Scenario: We only want the job title, salary, and final risk category
selected_cols <- ai_jobs %>%
  select(Job_Title, Average_Salary, Risk_Category)

print("--- Selected Specific Columns (Job_Title, Average_Salary, Risk_Category) ---")
print(head(selected_cols, 3))
```

```
# Method B: Select a range of adjacent columns
# Scenario: Select everything from 'Years_Experience' to 'AI_Exposure_Index'
range_cols <- ai_jobs %>%
  select(Years_Experience:AI_Exposure_Index)

print("--- Selected Range of Columns (Years_Experience to AI_Exposure_Index) ---")
print(head(range_cols, 3))

# Method C: Select using helper functions (e.g., starts_with)
```

```
# Scenario: Select columns that start with "S" (Skill_1, Skill_2, etc.)
starts_with_s <- ai_jobs %>%
  select(starts_with("S"))

print("--- Selected columns starting with 'S' (The 10 Skill columns) ---")
print(head(starts_with_s, 3))

# ------------------------------------------------------------------------------
# 3. DROPPING VARIABLES (Removing Columns)
# ------------------------------------------------------------------------------
```

```
# We use the minus sign (-) to remove variables

# Method A: Drop a single specific column
# Scenario: Remove the 'Tech_Growth_Factor'
dropped_one <- ai_jobs %>%
  select(-Tech_Growth_Factor)

print("--- Dataset with 'Tech_Growth_Factor' dropped (Names to verify) ---")
print(names(dropped_one))

# Method B: Drop multiple columns
```

```
 # Scenario: Remove 'Years_Experience'
and 'Education_Level'
dropped_multiple <- ai_jobs %>%
  select(-Years_Experience,
-Education_Level)

print("--- Dataset with
'Years_Experience' and
'Education_Level' dropped (Names to
verify) ---")
print(names(dropped_multiple))

# Method C: Drop a range of columns
# Scenario: Remove all 10 Skill columns
(from 'Skill_1' to 'Skill_10')
dropped_range <- ai_jobs %>%
  select(-(Skill_1:Skill_10))
```

```
print("--- Dataset with range 'Skill_1' to
'Skill_10' dropped (Names to verify) ---")
print(names(dropped_range))
```

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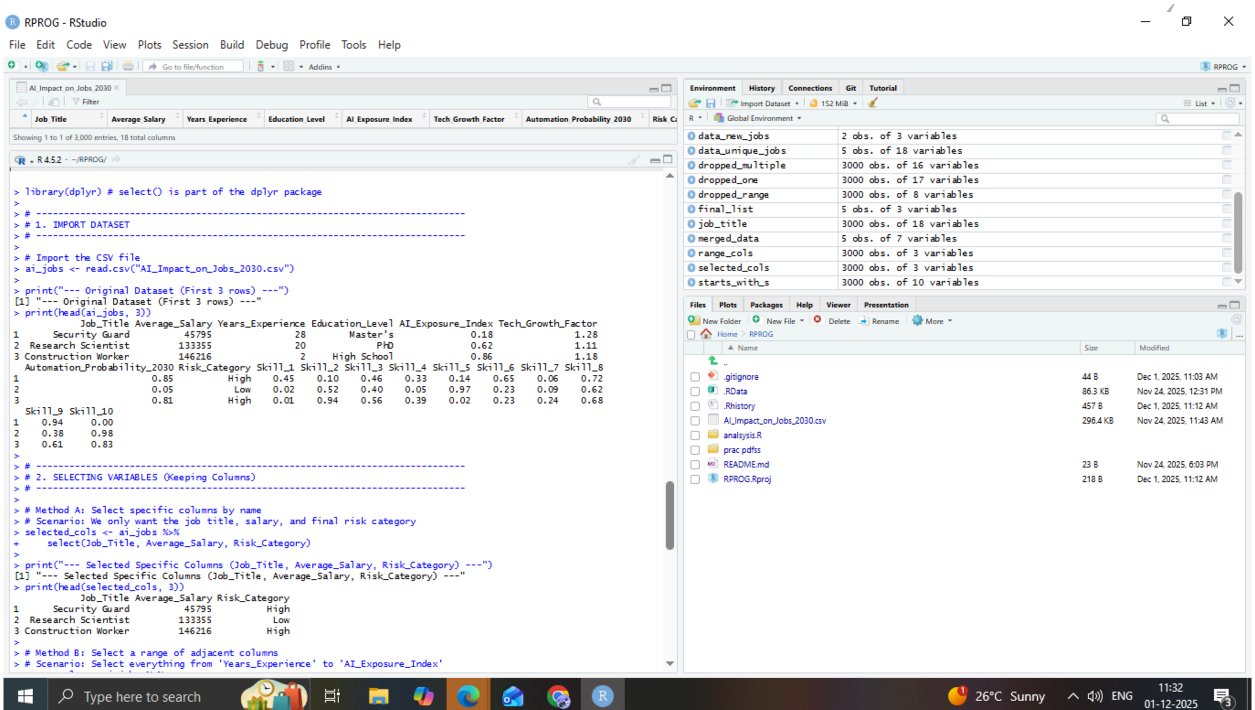

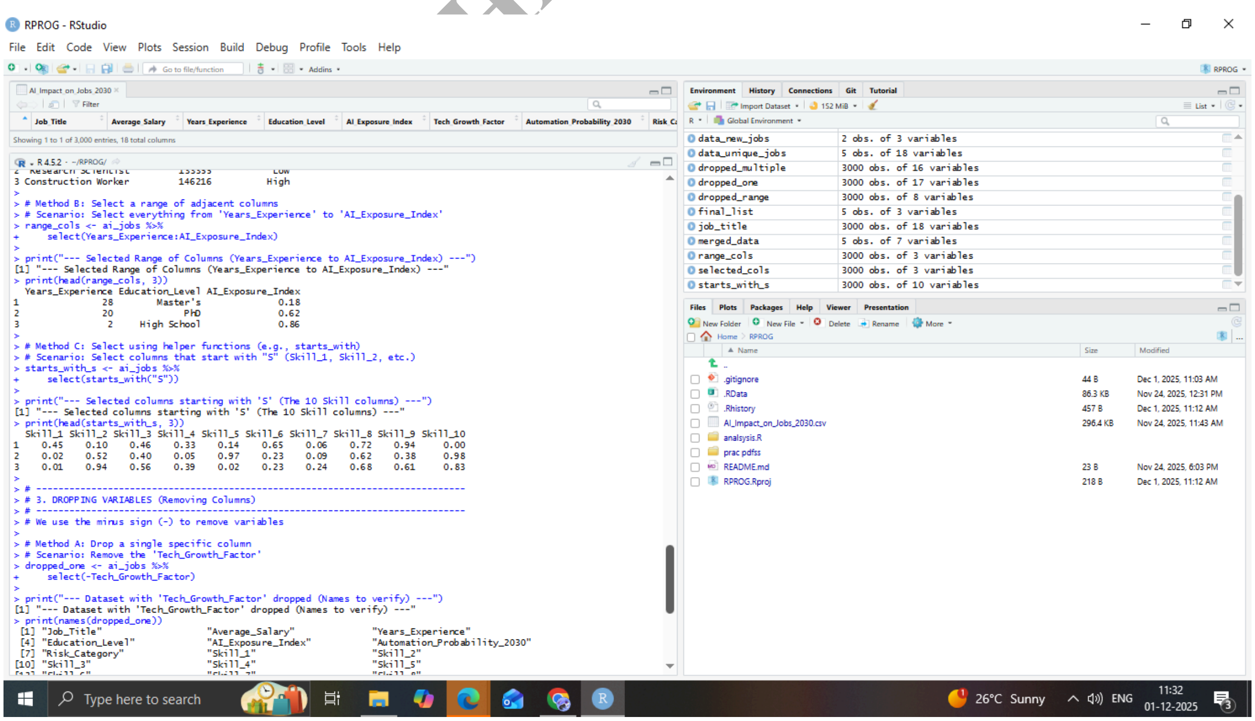

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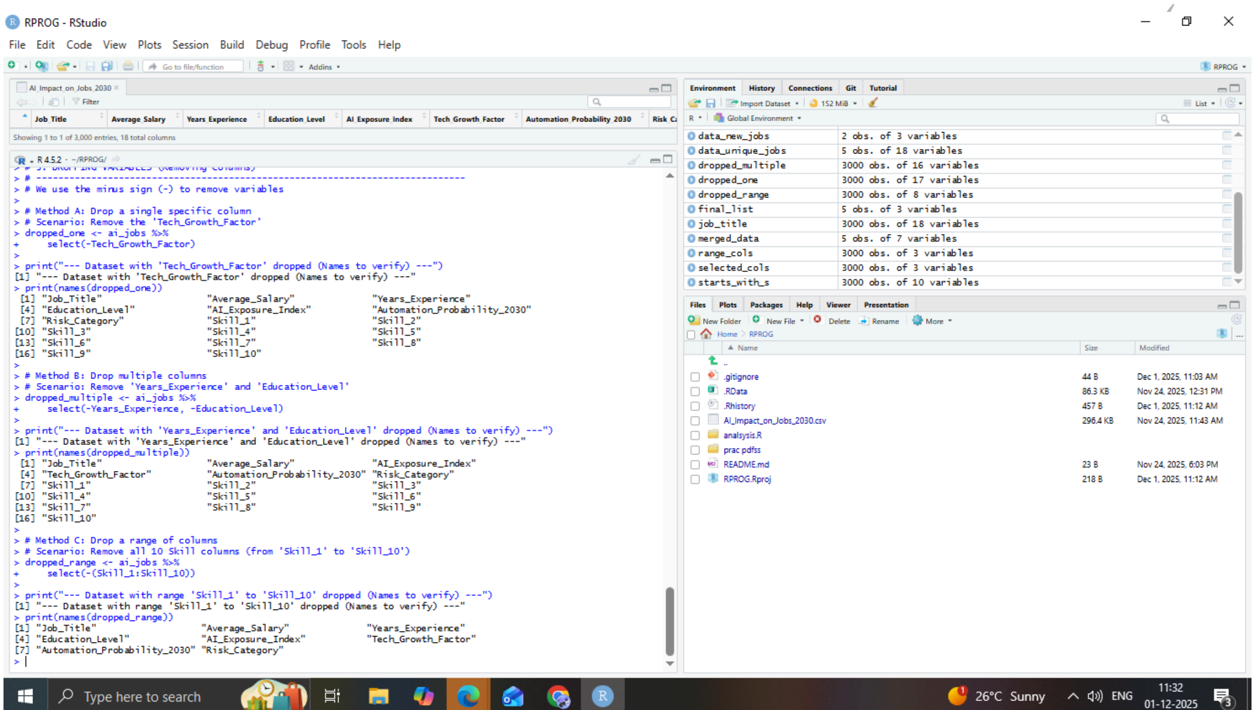

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